On Unconscious Influences

things that remain to do:

- [ ] improve discussion of Freud, & examples of his diagnoses
- [ ] replace guide-god example which gets the right predictions / explain how guide-dog examples leads to inconsistent choices
- [ ] discussion of why guide-dog model implies getting hypotheticals / wrong.
- [ ] add results of experiment demonstrating unconscious knowledge

Introduction

Over the last few years I've spent a lot of time in coffee shops looking out the window, thinking about decision-making & the unconscious, scribbling little nuggets in a notebook. I ended up writing two papers ("Hierarchical Aggregation of Information and Decision-Making", and "Implicit Preferences Inferred from Choice" with Jon de Quidt). The papers are fairly technical, and this post is meant to be a sort-of layperson's guide to the whole area, and an informal statement of the ideas.

I'm going to leave it to another post to make the case for the *magnitude* of unconscious influences – here I just try to organize a lot of different methods of identifying unconscious influences, and which ones seem right to me.

Here is the argument in a nutshell:

1. There are plenty of reasons to think that unconscious influences are strong – in other words, that people have limited insight into what factors influence their decisions.
2. The idea of unconscious influences has been in and out of the mainstream of psychology for the last 200 years, but always hounded by arguments over what it means, i.e. over what evidence would be sufficient to show that a decision was influenced by an unconscious factor. The battle has seen many reversals: many new types of evidence have been discovered which are thought to reveal unconscious influences, and then later the techniques or interpretations have been shown to have flaws and the line of inquiry fizzles out.

3. Two common sources of evidence: (A) people reveal their unconscious preoccupations in their involuntary responses – e.g. how their pupils dilate, how quickly they respond to a stimulus, in their word associations, dreams, slips of the tongue; (B) people reveal unconscious influences in discrepancies between how they act and how they explain their behaviour. Both sources of evidence have got tangled in debates about interpretation, and there are substantial camps on either side with not much agreement on what constitutes sufficient evidence for unconscious influences.
4. A third type of evidence is less common but, I think, more powerful: evidence from inconsistencies in decision-making. The idea being that unconscious factors are by their nature *isolated* from conscious factors, i.e. they don't interact with conscious beliefs and desires, and this isolation will cause characteristic inconsistencies.
5. This can be made precise with an analogy: the relationship between the conscious and unconscious brain is like the relationship between a blind man and his guide dog. The blind man makes decisions based, in part, on which way the guide dog is pulling, so the guide dog's beliefs and desires influence the man's decisions, but without the man knowing exactly what those beliefs and desires are, and so he couldn't tell you how much any particular factor contributed to his decision, because he doesn't know how much it contributed to the guide-dog's decision about which way to pull. Testing for unconscious influences in behaviour is just testing the degree to which your brain is being led by a guide dog.
6. This implies two ways of testing for unconscious influences: (1) testing whether people can accurately answer hypothetical questions about their own decisions; and (2) testing whether people make consistent judgments when judging two outcomes at a time.
7. First, hypothetical questions. We can ask people, how would your judgment change if this factor changed? Would you still like this painting if the name of the artist was different? Would this drawing look more like your cousin if the nostrils were bigger? Unconscious influences imply that people will not be able to give accurate answers to these hypothetical questions – they don't know what they would do until they are actually in the situation (AKA, they don't know which direction they would go in until they feel which direction the guide-dog is pulling).
8. The second way of testing for unconscious influences is the subject of my paper with Jon: we can observe unconscious influences in *joint* decisions – i.e. when we observe someone making two judgments simultaneously or consecutively (or choosing between two outcomes). When we observe more than one of these joint judgments we can identify inconsistencies between them. Suppose you had an unconscious preference for men over women, this can manifest as the following: when you see two CVs which are identical, except that one is a man and one is a woman, then you're indifferent between them; but when the two CVs differ in some other respect (e.g. one has a PhD, the other has an MBA), then you always have a preference for the CV belonging to the man. Your guide dog has a

bias towards men, which you're not aware of: the bias will only sway your decision in the second case because, in the second case, when your guide dog pulls you towards the man with a PhD, you don't know how much it's due to his masculinity, and how much due to his PhD.

9. In the end I think that our brains are full of guide dogs all pulling in different directions, often leading us in inconsistent circles. If we had the stomach for it we could plot out our decisions all on a map – and calibrate how each factor influences judgment.

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Motivating Examples

Here are some examples of possible unconscious influences, to help with being clear on what I'm talking about. I'll try to return to these at the end.

1. People judging a presidential candidate's actions differently depending on whether it's a man or a woman; judging an ethical lapse differently depending on whether committed by a Democrat or Republican; judging a study differently depending on whether it supports their own pet theory. In each case not being aware of that influence.

2. The little things that a family does to spite each other: someone says that Thursday doesn't suit them, in part because they know that Thursday suits you best (sometimes consciously, but also sometimes unconsciously).

3. Freudian conjectures about trauma causing later responses: e.g. Freud thought Anna O's aversion to water was because of a memory of seeing a dog drinking from a glass of water. More ambitiously, Freud's 'seduction' theory, that psychosomatic illness is due to repressed memories of childhood sexual abuse.

4. Your preferences changing with the circumstances: saying yes to a friend, a hairdresser, bartender, waiter, or salesperson, because it seemed like a good idea at the time, and later not understanding why you didn't say no.

5. Knowledge used in your judgments, but not consciously accessible: knowledge of grammar implicit in your judgements of whether a sentence sounds right; knowledge of the physical world implicit in your judgements of distance and velocity of the things you see; knowledge of faces implicit in recognizing your cousins.

My examples include both unconscious knowledge and unconscious desires. I'm not going to talk about unconscious *sensations* – flashes of advertisement that persuade me to drink Coca Cola – but I think you can also think about those in a similar way.

Ways of Measuring Unconscious Influences

Here are three ways of identifying whether an influence is conscious:

1. **Involuntary responses.** In the dilation of your pupils, in your response times, in slips of the tongue, what you see in Rorsach blots, in your word associations, in your dreams.
2. **Ability to describe the influence.** Ask people what they think influences their decisions, and compare that to what actually influences their decisions.
3. **Integration with other influences.** Finally we can say that a factor is unconscious if its influence seems to be partitioned off from other influences. The most simple example is if a factor continues to influence you even in cases where you have reason to ignore it.

Involuntary responses

Freud is the most famous theorist of extracting unconscious factors from involuntary responses – he wrote three books on different methods: one on dreams, one on jokes, one on mistakes (mis-reading, mis-hearing, mis-speaking). An example from the last book: "A woman who is very anxious to get children always reads 'storks' instead of 'stocks'." Most of Freud's examples of unconscious influences are more complex than this one, and more often the hidden factor is something unpleasant or shameful.

Another way of measuring unconscious cognition is through measuring arousal. Most famous is the "Iowa card task" first used by Bechara et al. (1996). They had their subjects choose among playing cards, and receive rewards if they chose certain cards. They found that people gradually learned which types of cards were rewarded, but they also found that the subjects' automatic responses (measured by skin conductance, i.e. sweating) would show an awareness of the pattern more quickly than the subjects' choices would: i.e., after a while, when the subject's hand hovered over a card which was rewarded, then the subject would sweat a little more, even though the subject wasn't any more likely to choose that card. They said this showed that unconscious learning was outpacing conscious learning. Antonio Damasio, one of the original authors, went on to write *Descartes' Error* – accusing Descartes' of starting the great misapprehension that emotions and reason are in competition. Damasio said that his experiments showed that, instead, emotions inform reason and improve

decision-making. A lot of subsequent papers have tried to show that snap decisions, which avoid conscious processing, can produce better outcomes than slow considered decisions.

Even more famous is the "Implicit Association Test" (IAT) (Greenwald, McGhee and Schwartz (1998)). Subjects are told to press a button whenever they see something from either of two different categories of stimuli, e.g. press the button if you see either a black face or a word with a positive association. Their finding, much-replicated, was that people are relatively quicker at the task (meaning they have shorter response times) when black faces are paired with negative words, or white faces are paired with positive words, than the other way around. They find that this occurs even among people who report no conscious negative feelings towards black people, and they interpret this as revealing an unconscious association between black people and negative feelings, and they argue that this association could affect your decision-making without you being aware of it.

Many other measures of automatic responses have been popular at different times: hypnosis and word association (Freud used both of these before moving to talking therapy); Rorsach blots (AKA inkblot tests); thematic apperception test (interpret an ambiguous drawing, still widely used); lie detectors (see how much you sweat when you answer a question).

Unfortunately a great deal of this research turns out to be both hard to replicate, and relying on strong assumptions to interpret the experimental evidence as reflecting unconscious associations. Newell and Shanks (2014) give a powerful arguments for both of these points.

However even if we had solid evidence that there are important unconscious influences on involuntary responses, they don't show that these reach all the way to decision-making. If unconscious associations only show up in sweating, response time, and dreams, but evaporate in decision-making, then unconscious associations are not very important for social science. Most of the authors in this literature have *assumed* that the unconscious factors they identify affect real decisions, but have left that extrapolation untested. Blanton et al. (2009) say that there's no persuasive evidence that implicit racial bias, as measured by the IAT, predicts peoples' decision-making, once you control for measures of *explicit* racial bias, i.e. when you just ask people how they feel about black people. (Singal (2016) has a long discussion).

Ability to describe the influence

A second type of evidence is to compare self-reported influences on behaviour with actual influences on behaviour. Here are some examples:

1. In the mid 20th-century behaviourists found that they could shape their subjects choices through conditioning with rewards and punishments, while the subjects seemed to remain ignorant of this shaping. For example if you

say 'mm-hmm' whenever someone uses a plural noun, then after a while that person ends up using plural nouns more often, apparently unaware of the influence (Thorndike and Rock (1934); Greenspoon (1955)).

2. Since the 1970s social psychologists have published all sorts of experiments in which they vary an apparently irrelevant factor and find that this can affect peoples' decision-making. Nisbett and Wilson (1977) summarize a lot of experiments and say "subjects are sometimes (a) unaware of the existence of a stimulus that importantly influenced a response, (b) unaware of the existence of the response, and (c) unaware that the stimulus has affected the response."
3. Another paradigm from the 1970s asks people to make a judgement - e.g. which stock to pick - and also rate the importance of factors which contributed to their decision. Slovic et al. (1972) find a correlation of only 0.34 between the implicit and explicit weights of 13 stockbrokers.
4. Finally, since the 1970s a smaller group of psychologists have been running experiments in which people learn a complicated pattern, and then are asked about their insight into it. E.g. in Arthur Reber's "artificial grammar" experiments subjects learn, through trial and error, to discriminate between two categories of words. After some time they become very good at the task, but when asked to explain how they are making decisions they are very bad at explaining rules that they seem to know.

Similarly to the previous category, a lot of this evidence is very fragile: either hard to replicate, or based on delicate interpretations of what is happening in the experiment. Newell and Shanks (2014) again give a good summary.

An additional problem is that it could be simply that verbalizing knowledge is hard. You may be consciously aware of something, but clumsy in expressing it. And this literature is full of reversals which bear this out: when experiments are repeated it has often turned out that the subjects *do* report awareness of the pattern that they have learned if they are asked the question in a different way. Mitchell et al. (2009) say "[i]t is very difficult to provide a satisfactory demonstration of unaware conditioning simply by showing conditioning in the absence of awareness. This is because it is very difficult to be sure that the awareness measure and the conditioning measure are equally sensitive."

Some more examples of describing influences: the "sugar factory" game; sequence learning; Mitchell on evaluative conditioning (p191); Perruchet's air puffs; expert judgment and stating the importance of different factors in influencing judgement.

Isolation of Unconscious Influences

Finally there's a third type of evidence which is more strictly behavioural: an unconscious factor is one which is *isolated* from your other conscious beliefs and desires – i.e. it does not interact with conscious factors – and that isolation will be reflected in your behaviour. This isolation criterion has been given various

names, but I don't think it's ever been explained as clearly as it could be.

[new dog example]

To be completely precise think of the blind man (the conscious part of the brain) and the guide dog (the unconscious). The guide dog can know something – e.g. she knows when the crossing light is flashing – which the man does not know, and her knowledge can influence the man's decisions. The guide dog's knowledge is isolated from the man's knowledge: it only influences his decisions through the guide dog's recommendation which is transmitted by pulling on the leash. Suppose you tell the man that the crossing lights are not working properly, and so any flashing is entirely at random and uninformative. The man and dog, considered as a system, has two pieces of information: (1) the light is flashing; and (2) the flashing is uninformative. However the first piece of information is isolated from the second, implying that they will not be integrated, because neither the man or dog knows both. This will be reflected in the man's behaviour: he will still be influenced by the guide-dog's recommendation, because the dog sees more things besides the crossing-light, and so his decision about crossing the road will be influenced by the light, even though he *knows* that the light is irrelevant. [Explain that this will produce inconsistencies in decision-making.]

I know of just a few cases where the isolation of knowledge has come up clearly in trying to define or measure unconscious influences.

Stich (1978) said that certain mental states are "inferentially unintegrated":

> "[unconscious beliefs are] largely inferentially isolated from the large body of inferentially integrated beliefs to which a subject has access"

He gives an example: suppose Chomsky has a theory of grammar, and that there exists some grammatical rule which is a counterexample to that theory. If a linguist knows that rule consciously, then the linguist will immediately infer that Chomsky's theory is false. But if the linguist only knows the rule *unconsciously*, then they won't be able to make that inference, because the knowledge is "inferentially unintegrated" – i.e. the knowledge is isolated from the knowledge regarding Chomsky's theory. [1]

A separate place where this separation has come up is in the work of Zenon Pylyshyn and Jerry Fodor since the 1980s regarding perception being "cognitively impenetrable," or "informationally encapsulated." They mean that perceptual processes often make inferences without taking into account all the information that is available, i.e. by drawing only on a subset of information. Their principal argument was from perceptual illusions: they argue that illusions can often be understood as rational inferences from a subset of the information available.

[1] Quine says you shouldn't call this type of thing unconscious knowledge – your linguistic practice may *obey* some rule, but you can't say that you unconsciously *know* that rule, because there are infinitely many different rules that would imply that pattern of behavior. But this skeptical objection is too tough: Quine would deny that a cow can have a belief about where a water trough is, & instead admit only that the cow's behavior is consistent with a particular belief among infinitely many others.

Helmholtz had a nice example: if you press with your finger on the edge of your eyeball then you'll tend to see a point of light appear, but on the *opposite* side of your field of vision from where your finger is. This is because the left side of your retina receives light from the right side of your visual field, and vice versa. So when your eye receives some stimulation on the left-hand side, something in your brain makes an inference that light is coming from the right-hand side. This is a sensible inference from your eye's point of view. Even though you are consciously aware that the true cause of the stimulation is a finger, your visual cortex interprets the stimulation as light, coming from the opposite side, because it isn't party to the information that is available to the conscious brain.

This isn't quite the same as the case of the blind man and the guide dog. In the example above the pre-conscious process has a *subset* of the information available to the conscious brain. I.e., the man doesn't need to be blind in this metaphor – it's just a case where the dog leads in a different direction than the man would. Fodor and Pylyshyn were mostly interested in cases where the pre-conscious processing works with a strict *subset* of the information available to the conscious brain – although Fodor (1983) does have a brief discussion on whether early perceptual processes have access to information not available to the conscious brain, which would imply unconscious influences, in my sense.

Finally the isolation argument has appeared in the literature on human "associative learning," and whether or not the associations that we learn through conditioning are unconscious. A typical experiment involves ringing a bell and giving an electric shock. After a while people learn to flinch when they hear the bell. For a long time psychologists tried to map out the logic of how such associations would form, trying to figure out the rule which governed learning of associations. However in the last few decades an argument has been made that these learned associations are not in fact mechanical - there is no simple rule - instead they are more-or-less optimal responses to the environment based on the entirety of the information available, i.e. they are not *isolated* from other knowledge (though the argument isn't usually put in terms of conscious vs unconscious knowledge). For example Colgan (1970) told subjects, after they learned an association, that the association is no longer valid ("from now on the bell will not signal an electric shock") and he found that, although this didn't entirely extinguish the flinching, it did cause it to markedly decrease. So the flinching is not isolated from your conscious knowledge: the association, at least to some degree, interacts with more abstract knowledge. There are many other circumstances where rule-based theories of association-learning have foundered because it turns out that peoples' responses respond to outside considerations. De Houwer, Vandorpe and Beckers (2005) say > The two types of models can be differentiated ... by manipulating variables that influence the likelihood that people will reason in a certain manner but that should have no impact on the operation of the associative model. We have seen that such variables (e.g., instructions, secondary tasks, ceiling effects, nature of the cues and outcomes) do indeed have a huge effect. Given these results, it is justified to entertain the belief that participants are using controlled processes such as reasoning and to

look for new ways to model and understand these processes.

Mitchell says:

> "The results consistently show evidence for skin conductance [effects] only in participants who are aware of the [relationship] . . . [a]lthough there are many papers arguing for unaware conditioning, close inspection reveals, in almost all cases, that the measure of conditioning was most likely more sensitive than that of awareness."

In retrospect a lot of behavior that was studied in the lab, which was thought to be telling us about the wiring of the animals, actually was telling us about the world outside the animal, because it has turned out that the animals' response is the *optimal* response to the typical circumstances it faces in the world. (See also Mitchell et al. (2009) section 4.3.)

If this line of thought were entirely correct – if all information was integrated and fed into every decision – then there would be no unconscious influences in my sense. However I do think that there's plenty of evidence that remains for a lack of integratation between cognitive processes.

Summary so far

1. Looking at involuntary reactions tells us something about your reactions to a situation, but it doesn't tell us how big the influences are on your decisions.
2. Comparing peoples' decisions to what they say about their decisions is also imperfect evidence for unconscious associations, because people can struggle to explain their decisions without being ignorant of the reasons behind your decisions.
3. Finally, internal coherence of decision-making seems a much more solid way of identifying unconscious influences.

Guide Dogs in Your Head

My paper, "Hierarchical Aggregation of Information and Decision-Making", essentially spells out the guide-dog model of unconscious influences, in technical detail. It has a few implications which can all be explained in terms of the guide-dog metaphor:

1. You can't answer hypothetical questions accurately. In the same way a blind man wouldn't know what he'd do in another situation, without hearing from his guide dog.
2. You make inconsistent comparisons. .

Difficulty Introspecting

I'm going to say that there are unconscious influences on my decisions if I can't accurately predict how I would act if the circumstances were different. That shows that I'm not conscious of how different factors contribute to my decisions.

This definition includes both cases where (A) I *think* I know what affects my decisions, and I'm wrong; or (B) I know that I don't know what affects my decisions.

We could then put the question a different way, instead of "how important are unconscious factors?" say "how deep does conscious thought go?" I.e., how much do we understand of our own motivations and reasoning?

Why would we assume that it goes all the way down? Animals make a lot of sensible decisions but probably don't have much insight into the interplay of reasons. We certainly have limited conscious insight into our *perceptual* judgments – I couldn't explain what it is about my cousin that makes me recognize her, or what it is about the trajectory of a ball that makes me think it's going to land in a particular place.

To ask whether you know motivations is a pretty woolly question – much more precise is whether people can give accurate answers to hypothetical questions: what would you think or do in this situation? This gives a concrete meaning to unconscious influences.

Overall I think that people are *bad* at this. I.e., they .

Here, first, are some shallow hypotheticals, which I think I can give roughly accurate answers to (but I'd still not be surprised if I was wrong):

- Would I still run for the bus if I had 5 minutes, instead of 6?
- Would I buy this chicken if it were $6/pound, instead of $5?
- Would I have chosen to buy fish, if the chicken wasn't available?

But there are also a lot of deeper ones, which I find pretty difficult to answer.

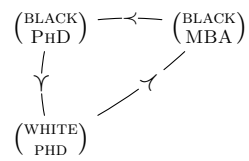
- Would I still run for the bus if I never saw other people running for the bus?
- Would I behave the same way towards colleague X if they were a woman instead of a man?
- What would I think of my parents if I was to meet them now, as strangers?
- What would I think of my work if I came across it as someone else's?
- Would my politics be the same if I went to a PhD program at a different school (where the professors' politics are different)?
- Would I feel the same way about my favorite album if they were selling it in Starbucks?

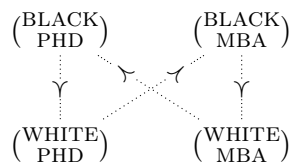
Bundling Factors: Triangles and Figure-Eights

Jon and my paper builds on this basic idea – that when factors are *bundled*, then the unconscious influence will tend to increase.

Here we need to make a fine distinction: suppose an unconscious factor x has a weight w in your judgment.

Here are a collection of inconsistent choices that a person could make, each of them revealing an unconscious preference for White over Black:





$$y\left(BP, \begin{bmatrix} BP & BM \\ & \end{bmatrix}\right) < y\left(BP, \begin{bmatrix} BP & \\ & WM \end{bmatrix}\right)$$

(couple of diagrams missing, need to fill these in.)

$$y(\mathrm{B}, \{\mathrm{B}\}) < y(\mathrm{B}, \{\mathrm{B}, \mathrm{W}\}) y(\mathrm{W}, \{\mathrm{W}\}) > y(\mathrm{W}, \{\mathrm{B}, \mathrm{W}\})$$

Mapping out unconscious influences

So how should we go about testing for the existence of, and mapping out, unconscious influences?

Think of your responses to different outcomes as a landscape of hills (positive responses) and valleys (negative responses).

Returning to the Motivating Examples

- **People judging a presidential candidate's actions differently depending on whether it's a man or a woman; people judging an ethical lapse differently depending on whether committed by a Democrat or Republican; academics judging a study differently depending on whether it supports their own pet theory.**
- **The little things that a family does to unconsciously spite each other: someone says that Thursday doesn't suit them, in part because they know that Thursday suits you best.**

Here the factor influencing you unconsciously is whether your family-member is free on Thursday.

- Judging hypotheticals:
- Judging jointly:
- **All the Freudian literature on repressed desires and childhood traumas manifesting themselves in adult decision-making. Maslow: "A desire for an ice cream cone might actually be an indirect expression of a desire for love."**

Suppose the factor unconsciously influencing you is your father's coldness when you were a child, and this causes you to be especially sensitive to praise by your boss. This is difficult to apply our two methods to, because it's hard to entertain counterfactuals: what would you do if your father were warmer? How would we know the correct answer?

I guess in principle one could do this: (1) ask people whether they believe that, on average, cold fathers cause people to be more sensitive to praise; (2) do the study, and find out whether cold fathers really do have this effect. This would show that, on average, people are not aware of a factor that has an influence on their decisions.

- **Your preferences changing with the circumstances: saying yes to a friend, a hairdresser, bartender, waiter, or salesperson, because it seemed like a good idea at the time, and later wondering why you didn't say no.**

** **Knowledge used in your judgments, but not consciously accessible: knowledge of grammar implicit in your judgements of whether a sentence sounds right; knowledge of physics implicit in your judgements of distance and velocity of the things you see; knowledge of faces implicit in recognizing your friends.**

As an example, take the order of adjectives. The phrase "big red car" sounds better than "red big car." However it's hard to explain why. Let's say that the rule is that X adjectives always precede adjectives of type Y.

Then judging hypotheticals can be run this way: give someone a list of adjectives and an incomplete sentence, e.g. "big ____ car." Then ask them to choose which of the adjectives would form a grammatical sentence. If they can only answer accurately by trying out each adjective, one by one, in the sentence, then their knowledge of grammar is unconscious.

Conclusion

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Odds & ends

When the guide dog has different preferences from you

My model of unconscious influences assumes that the final decision is always made by the blind man, and the dog just gives advice. However it's also interesting to think about when the guide dog can pull so hard that she forces the man to follow her – i.e. mental factors which have some influence over the decision. There are a lot of papers in economics on multiple-selves models of decision-making which are like this – the different selves have different preferences. There's a less-interesting and more-interesting version of this idea: if the two selves differ *only* in preferences, then it's not really different from a single self who has multiple competing preferences. It's more interesting if the two selves have access to different sets of information. Then there's a simple implication: if your guide dog has a preference that you don't share, then you'll take care in what information you expose them to – you won't walk your dog past the butcher's shop. The papers in economics mostly talk about this kind of result, they try to rationalize all sorts of economic puzzles as people managing their guide dogs. (The future-self vs present-self tradeoff can be thought of in guide-dog terms: the guide-dog only sees the present costs and benefits of an action, not the future consequences). All of this fits within the Fodor-Pylyshyn setup where the dog has a subset of all available information. I think very little has been done in thinking about the degree to which the guide-dog has information that the man doesn't have – i.e. when they differ in both preferences and beliefs. Then it becomes a little like a cheap-talk or persuasion game: the guide dog is giving you advice, and you know that they have a particular bias, yet you cannot completely was that bias out.

History of Implicit Preferences

The existence of the unconscious is one of the few modern philosophical ideas, i.e. one of the few not already discussed by the ancient Greeks.

In 1817 in *Mansfield Park* Jane Austen wrote about Thomas Bertram perceiving that a rich suitor was showing interest in his niece, and that he could not "refrain (though unconsciously) from giving a more willing assent to invitations on that account'." Austen had used the word "unsconscious" before that, but only to mean that people were doing something unintentionally, not that they were unaware that some factor was influencing their judgment. In the same year, 1817, Coleridge first used the term "the unconscious" in English (imitating Schiller's use in German).

It's an interesting question why people didn't talk about unconscious influences prior to the 18th century, given that it's been such a popular thing to talk about ever since. They weren't stupid – Plato, Augustine, & Shakespeare weren't naive about human nature – if unconscious influences are a big deal, why didn't they talk about them? If it was there all along, why did it take us so long to discover it?

Perhaps this answer is too cute, but you could say that the unconscious is only a problem for people who expect to be able to give a reason for everything they do. Let's say that human motivation is an emulsion – drops of logic suspended in a liquid instinct. We often make little logical jumps, like preferring two dollars to one dollar. But more often we have to consult our instincts: when we have to choose between a kiss and a dollar, logic doesn't help, and we don't have a schedule of preferences we can consciously consult to decide whether it's a good deal. Only in the 18th century did European intellectuals begin to expect that we should find an entire logical skeleton underneath our decisions and give an explicit reason for every choice, and perhaps our failure in being able to explain ourselves led to the practice of talking about the unconscious as a separate mental process.

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Hart Blanton, James Jaccard, Jonathan Klick, Barbara Mellers, Gregory Mitchell, and Philip E Tetlock (2009). "Strong claims and weak evidence: reassessing the predictive validity of the IAT". Journal of Applied Psychology, 94(3): 567.

Greenwald McGhee & Schwartz (1998) "Measuring individual differences in implicit cognition: The Implicit Association Test."

De Houwer, Vandorpe and Beckers (2005) On the role of controlled cognitive processes in human associative learning * "In hindsight, it seems obvious that people can learn about associations by using controlled processes such as reasoning and hypothesis testing. Why then, are associative models still dominant in modern research? One reason is that the associationistic view has a long tradition in psychology (and philosophy). It is thus difficult for many people to leave behind the associationistic view that has guided their thinking and research for many years. Another important reason is that associative models do quite well in accounting for the available empirical data. The well known Rescorla-Wagner model (Rescorla & Wagner, 1972), for instance, is compatible with a huge number of findings while being relatively simple. If our argument is correct that associative models do not provide an accurate account of the processes that underlie associative learning, how is it possible that they are able to account for so much of the data? We agree with Lovibond (2003, p. 105) that"the success of these models is due to them capturing, at least in part, the operating characteristics of the inferential learning system". What this means is that associative models (as well as probabilistic models for that matter) can be seen as (mathematical) formalisations of certain deductive reasoning processes. A system that operates on the basis of associative models does not reason, but acts very much as if it is reasoning. The associative models will thus often predict the same result as a model that is based on the assumption that humans actually generate and test hypothesis or reason in a controlled, conscious manner. The two types of models can be differentiated, however, by manipulating variables that influence the likelihood that people will reason in a certain manner but that should have no impact on the operation of the associative model. We have seen that such variables (e.g., instructions, secondary tasks, ceiling effects, nature of the cues and outcomes) do indeed have a huge effect. Given these results, it is justified to entertain the belief that participants are using controlled processes such as reasoning and to look for new ways to model and understand these processes."

Newell & Shanks (2014, BBS) "Unconscious Influences on Decision-Making: A Critical Review"

- "There is little convincing evidence of unconscious influences on decision making in the areas we review" – they say many of Nisbett and Wilson's results on lack of introspection have been reinterpreted
- (1) multiple-cue judgment : classic paradigm is that you compare weights revealed by decisions to self-reported weights. They say that protocol for getting self-reported weights are usually shitty, & better measurements show that people understand their own judgment better
- (2) decisions are better when you don't deliberate (all the evidence for this is shitty)

- (3) awareness / Iowa gambling task (your subconscious is attracted to the best deck, but conscious is not): very shitty evidence
- (4) priming: people often can report what primes were presented, if you try harder- they say that there is strong evidence for top-down influence even in perceptual tasks. [I emailed them suggesting the joint/separate randomization task, they said it sounds nice]
- They say that dispute over definitions is over: "A surprising outcome of the review is that debates and disagreements about the meaning of the terms consciousness and awareness have (with a few exceptions) played a remarkably minor role in recent research. Whereas issues about how to define and measure awareness were once highly prominent and controversial (e.g., Campion et al. 1983; Reingold & Merikle 1988), it now seems to be generally accepted that awareness should be operationally defined as reportable knowledge, and that such knowledge can only be evaluated by careful and thorough probing."

Michael Polanyi (1966) "The Tacit Dimension"

Nisbett & Wilson (1977) "Telling More than we can know" A very influential paper (7000 citations) arguing that unconscious influences are rampant: "Subjects are sometimes (a) unaware of the existence of a stimulus that importantly influenced a response, (b) unaware of the existence of the response, and (c) unaware that the stimulus has affected the response." However a lot of the experimental literature in this field is now under a cloud – either through being hard to replicate, or being based on arguable interpretations.

Quine (1972) "Methodological reflections on current linguistic theory" When Quine wrote this, linguists were talking a lot about how people had unconscious knowledge of grammatical rules. Quine said that you can't say that someone knows a rule, from observing that their behaviour conforms to that rule, because there will always be infinitely many different rules which the behaviour satisfies. So this is Quine pouring some cold water on unconscious knowledge. However, in response, I say that we can still define unconscious knowledge underneath behaviour as the inability to accurately answer hypotheticals. So we don't need to specify *which* rule the person is using, but it's clear that they do not have conscious access to any rule which generates their observed behaviour.

Stich (1976) "Beliefs & Subdoxastic States"

Suzanne O'Sullivan (2015) "It's All in Your Head: Stories from the Frontline of Psychosomatic Illness"

> p191: "For all the shortcomings in the concepts proposed by Freud and Breuer in *Studies*, the twenty-first century has brought no great advances to a better understanding of the mechanism for this disorder. The terms dissociation and conversion are still widely in use ... the general principles of modern dissociation are not very different to

those of Victorian times ... in day-to-day practice Janet's and Freud's theories are regularly used, or misused."

Singal (2016) in New York Magazine - article on Implicit Association Test.